Why Regulators Will Soon Demand "Model Lineage" for Payment AI — And What Banks Must Build Now

By Neeraj Aggarwal

Executive Summary:
As AI becomes the invisible engine behind fraud detection, sanctions screening, routing, and risk decisions, regulators are confronting a new challenge: banks can no longer explain how their payment decisions were made. The next regulatory frontier will be **model lineage**—the ability to trace every AI-driven payment decision back through the exact chain of models, data, features, and transformations that produced it. Banks that cannot produce model lineage on demand will face compliance failures, audit gaps, and systemic risk exposure.

AI Is Now Making Payment Decisions Faster Than Banks Can Explain Them

Payment systems have quietly crossed a threshold.
Where once deterministic rules governed approvals and declines, today's payment flows rely on:

- Multi-layered AI fraud models
- Real-time behavioral analytics
- Sanctions-screening models
- Device-risk and merchant-risk scoring
- Routing optimisation engines
- LLM-based decision support

A single transaction may pass through **six to ten AI models** before a final decision is made.

Yet most banks cannot answer a basic regulatory question:

"Which model made this decision, using which data, at which version, under which conditions?"

This is the core problem regulators are preparing to address.

What “Model Lineage” Means — And Why It’s Different from Governance

Model lineage is not documentation.
It is not model governance.
It is not model risk management.

Model lineage is **forensic traceability**.

It provides a complete, auditable record of:

1. Model Version Lineage

- Model version
- Training dataset
- Training date
- Hyperparameters
- Drift metrics
- Approval history

2. Data Lineage

- Raw data sources
- Feature transformations
- Real-time enrichment
- Data quality scores

3. Decision Lineage

- Model outputs
- Thresholds applied
- Overrides
- Human-in-the-loop interventions
- Confidence scores

4. Explainability Lineage

- SHAP/LIME outputs
- Graph-based reasoning
- LLM reasoning traces
- Feature importance

5. Governance Lineage

- Policies in effect
- Regulatory mappings
- Audit trails

This is the level of transparency regulators will require for AI-driven payments.

Why Regulators Will Demand Model Lineage

The regulatory signals are already visible.

EU AI Act

Classifies fraud detection, AML, and credit scoring as **high-risk**, requiring traceability and explainability.

US Treasury, OCC, and Federal Reserve

Increasing scrutiny of AI-driven fraud and sanctions systems, especially in real-time payment environments.

FATF

Requires explainability for AML/CTF decisions.

Basel Committee

Expanding expectations for model risk management in AI systems.

FedNow and RTP

Real-time rails require **real-time accountability**.

The logic is simple:

> **If AI is making high-risk financial decisions, regulators must be able to audit those decisions.**

Model lineage is the only mechanism that provides this.

The Risks of Not Having Model Lineage

Banks without model lineage face increasing exposure:

1. Unexplainable False Declines

Leading to customer harm and regulatory scrutiny.

2. AI-Driven Fraud Loops

Where models reinforce incorrect patterns.

3. Sanctions Violations

Because the bank cannot prove why a transaction was cleared.

4. Routing Errors and Liquidity Shocks

Particularly in real-time payment systems.

5. Model Drift

Causing unpredictable behaviour in production.

6. Regulatory Non-Compliance

Resulting in enforcement actions and mandated remediation.

In the AI era, **lack of lineage equals lack of control**.

What Banks Must Build Now: A Model Lineage Architecture

Banks need a new architectural layer: a **Model Lineage Architecture** that sits alongside fraud systems, payment engines, and AI governance frameworks.

1. Data Lineage Layer

Tracks every dataset, feature, and transformation used in real-time decisions.

2. Model Registry Layer

A single source of truth for all models, including versioning, metadata, training history, and drift monitoring.

3. Decision Trace Layer

Captures every inference event, including inputs, outputs, thresholds, overrides, and confidence scores.

4. Explainability Layer

Generates human-readable explanations for regulators, auditors, and internal risk teams.

5. Governance Layer

Maps decisions to policies, regulatory requirements, and audit trails.

This is where banks can integrate modern governance frameworks such as **AI risk controls, resilience maturity models, and decision-audit pipelines**.

The Future: Real-Time Explainability for Real-Time Payments

Real-time payments require **real-time explainability**.

Within the next 24–36 months, regulators will expect:

- **Explainability APIs**
- **On-demand lineage reports**
- **Model-version tracebacks**
- **Real-time auditability**
- **AI governance embedded directly into payment flows**

Payment systems will evolve from:

Black-box AI → Transparent, auditable AI ecosystems

Banks that build model lineage now will be prepared for the regulatory shift.
Banks that wait will face compliance gaps, operational risk, and systemic exposure.

Conclusion

AI is transforming payment systems faster than regulatory frameworks can adapt.
But the direction of travel is clear:

If AI makes the decision, the bank must be able to explain it.
If the bank cannot explain it, it cannot defend it.
If it cannot defend it, it cannot operate it.

Model lineage will become the foundation of AI-driven payments.
The institutions that invest now will define the next decade of financial modernization.